Water Disinfection for Travelers

For Health Care Providers

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PURPOSE

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Top takeaway: Healthcare professionals should advise international travelers visiting regions with unsafe drinking water about effective water disinfection techniques.

Introduction

Waterborne diseases are a risk for international travelers who visit countries where access to safe water, adequate sanitation, and proper hygiene is limited, and

for wilderness visitors who rely on surface water in any country, including the United States. In both high-income and low- and middle-income countries, lack of safe drinking water is among the most immediate public health problems faced after natural disasters or in refugee camps. The list of potential waterborne pathogens includes bacteria, viruses, protozoa, and parasitic helminths.

Most of the organisms that cause travelers' diarrhea can be waterborne as well as foodborne. Many types of bacteria and viruses can cause intestinal (i.e., enteric) infection through drinking water (see Post-Travel Diarrhea). Common waterborne protozoa include *Cryptosporidium*, *Entamoeba histolytica* (the cause of amebic dysentery), and *Giardia*. Parasitic worms are not commonly transmitted through drinking water, but drinking water is a potential means of transmission for some.

International travelers may have no reliable resources to evaluate

local water system quality. While substantial progress has been made in the past 20 years toward the goal of safe drinking water and sanitation worldwide, according to the World Health Organization/United Nations International Children's Emergency Fund (WHO/UNICEF), in 2021, 25% of the world population still lacks safely managed drinking water in their homes and 1.7 billion people lack access to basic sanitation services; this can result in high levels of microbes in the environment and water sources. Where treated tap water is available, aging or inadequate water treatment infrastructure might not effectively disinfect water or maintain water quality during distribution. Where untreated surface or well water is used, and no sanitation infrastructure exists, the risk for waterborne infection is high.

All international travelers, especially long-term travelers and expatriates, should use one of the methods described in this chapter to ensure safe drinking water

when using an untreated or questionably treated water source. Bottled water has become the convenient solution for most travelers, but in some places, bottled water might not be superior to tap water (see Food and Water Precautions for Travelers). Moreover, plastic bottles create an ecological problem.

Water treatment and disinfection methods that can be applied in the field include heat, clarification, filtration, chemical disinfection, and ultraviolet (UV) irradiation. Several of these methods are scalable, and some can be improvised from local resources, allowing adaptation to disaster relief and refugee situations. Table 1.8.1 compares the advantages and disadvantages of the different methods. Additional information on water treatment and disinfection methods can be found at CDC's Water Treatment Options when Hiking, Camping, or Traveling website.

Table 1.8.1: Comparison of water treatment and disinfection techniques

	Advantages	**Disadvantages**
	• Does not impart additio	• Fuel source may be scar

	• Single step that is natural taste, odor, or color	ce, expensive, or unavailable • Does not imp

	• Effecti nactivatesallenteric pathogens	rovetaste, smell, orappearanceofsource

	venessis not compromised by contaminants	water • Does not prevent recontamination du

	orparticlesinthewater	ringstorage
	• Simpletooperate	• Somefiltertypes

	• Requires nohold ing time for treatment • Man	may be bulky or heavy to carry • Many filters d

	• Add sn oun ple y comm erci al pro duct desi gns	• More expensi ve vir uses not reli ably rem ove

	asantt astteorodor; oftenimprovestastea	vethanchemicaltreatment • Filtersseventu

	• Can be combined with and appear anywhere of water	all lycl og from sus pended part icul ate mat t

	chemicaldisinfectiontoincreasemicrob	er, requiringreplacement, maintenance,

	eremoval	orrepair • Doesnotpreventrecontamination

		nduringstorage
	• Inexpensiveandwidelya	• Impartstaste,odor;iod

	available in liquid or table t form • Result in	in eal so impart s color to water • Flexibled o

	gtastecanberemovedbysimpletechniques	singcanbecomplicatedtounderstandlodi

	• Flexible dosing • Equally easy to treat larg	neis physiologically active, with potent

	• Can provide short-term and small volumes	• Not readily effective

	disinfectant resistant for stored water • Ch	against *Cryptosporidium* oocysts • Effecti

	lorinedioxideismorepotentthananequival	venessdecreasesswithcloudywater • Liquid

	entdosesofchlorineandeffectiveagains	sarecorrosiveandstaincloothing • Chlorin

	tallwaterborne pathogens, including *Cry*	edioxidetabletsanddropsrequirepreparation

	p *t* *o* *s* *p* *o* *r* *i* *d* *i* *u* *m*	g e d c o n t a c t t i m e o f s e v e r a l h o u r s a n d h a v e n

		oprolongedresidualconcentration
	• Impa	• Requ

	• Portable battery-operated osteoporosis assessment system • Portable battery-operated osteoporosis assessment system	• Does not improve osteoporosis assessment accuracy • Does not improve osteoporosis assessment accuracy

	ated devices now available • Effective against	ppearance of water • Relatively expensive (

	nstallwaterbornepathogens • Extradoseso	exceptsolardisinfection[SODIS]) • Requi

	fUVradiationcanbeusedforaddedassuran	resbatteriesorpowersource(exceptSODl

	ceandwithnosideeffects	S) • Cannotknowifdevicesarredelivering re

		quiredUVdoses • Nopersistentresidualcon

		centration, sodoesnotpreventrecontami

		nationdurings torage

Field techniques for water treatment and disinfection

Heat

Common intestinal pathogens are readily inactivated by heat. Microorganisms are killed in a shorter time at higher temperatures, but temperatures as low as 60°C (140°F) are effective when a longer contact time (30 min) is used.

Pasteurization uses this principle to kill food-borne enteric pathogens and spoilage-causing organisms at temperatures between 60°C (140°F) and 70°C (158°F), well below the boiling point of water (100°C; 212°F).

Boiling is not necessary to kill common intestinal pathogens, but boiling is the only easily recognizable end point that does not require a thermometer. All organisms that may cause illness from drinking water are killed within seconds at boiling temperature. In addition, the time required to heat water from 60°C (140°F) to boiling works toward heat disinfection. Any water brought to a boil should be adequately disinfected; however, CDC recommends that travelers boil water for a full minute to account for user variability in identifying boiling points and to add a margin of safety. If fuel is scarce, heat water until first sign of simmering (small bubbles rising from bottom), reduce or remove heat, and leave the container covered for 30 minutes.

Although the boiling point for water decreases with increasing elevation, at common travel elevations the temperature needed to achieve boiling is still well above the temperature required to inactivate enteric pathogens. For example, at approximately 4,900 m (16,000 ft) elevation, the boiling temperature of water is approximately 83°C (182°F). In hot climates with sunshine, a water container placed in a simple reflective solar oven can reach pasteurization temperatures of approximately 65°C (150°F).

Travelers with access to electricity can bring a small electric heating coil, and many hotels have electric water pots to brew tea or coffee. When possible, travelers should avoid using water from the hot water tap for drinking or food preparation because hot tap water can contain higher levels of metals like copper and lead that can leach from water heaters and pipes.

Clarification

Clarification refers to techniques that reduce the cloudiness (turbidity) of water caused by the presence of natural organic and inorganic material. Clarification can markedly improve both the appearance and taste of the water. Decreasing turbidity reduces microbiological contamination but not enough to ensure potable water. Clarification techniques also facilitate filtration or disinfection by chemical treatment.

Coagulation and flocculation (C-F)

Large particles like silt and sand will settle by gravity (i.e., sedimentation). Cloudiness due to dissolved substances or smaller particles that remain suspended in water can be improved by using chemical products that coagulate and flocculate (i.e., cause clumping), allowing the solids to settle to the bottom of the water container and to be more easily removed by filtration or by carefully pouring

clear water off the top. This process removes many, but not all, microorganisms.

Alum, an aluminum salt widely used in food, cosmetic, and medical applications, is the principal agent for C-F. Travelers should add 1.3 mL (one-fourth tsp) of alum powder to 0.95 L (1 qt) of cloudy water; stir frequently for a few minutes and add more powder as necessary until clumps form. Allow the clumped material to settle into the bottom of the container or rise to the top and then pour the water through a coffee filter or clean tight-weave cloth to remove the sediment. Since most, but not all, microbes are removed, travelers must use a second treatment step (e.g., filtration, chemical disinfection, or ultraviolet). Some commercially available tablets or powder packets combine a C-F agent with a chlorine chemical disinfectant.

Filtration

Portable hand-pump or gravity-drip filters with various designs

and types of filter media are available widely. Filter pore size is the primary determinant of a filter's effectiveness (Figure 1.8.1). Manufacturers claiming a U.S. Environmental Protection Agency (EPA) designation of water "purifier" for their products must conduct their own testing to demonstrate their filters can remove at least 99.9999% of bacteria, 99.99% of viruses, and 99.9% of *Cryptosporidium* oocysts or *Giardia* cysts. The EPA does not independently test the validity of these claims.

Filter pore size

Most portable filters are microfilters with a pore size <1 µm, which should readily remove bacteria and protozoan parasites like *Cryptosporidium* and *Giardia*. Portable microfilters do not reliably remove enteric viruses (e.g., norovirus) that have an average size of 0.03 µm (Table 1.8.2). For areas with high levels of human activity in the watershed or in places with poor sanitation, travelers should use

finer levels of filtration or other techniques to remove viruses.

Table 1.8.2: Waterborne pathogens (average sizes) and filter pore size needed for their removal

If using a microfilter, travelers can treat water with chlorine to remove viruses. Progressively finer levels of filtration, known as ultrafiltration, nanofiltration, and reverse osmosis, all can remove viruses (Figure 1.8.1). Ultrafilters with pore size of 0.01 µm should be effective for removing viruses, bacteria, and parasites. Small portable ultrafilters are commercially available that use hollow-fiber technology and operate by gravity, hand-pump, or drink-through methods; however, these filters are both more costly and require greater pressures to push water through the filter, often at a slower rate.

Figure 1.8.1: Water contaminants, particle sizes and filtration methods

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Figure 1.8.1: Water contaminants, particle sizes and filtration methods

SOURCE: Backer H. Field water disinfection. In, Auerbach, PS, Cushing, TA, & Harris, NS, editors (2017). *Auerbach's wilderness medicine* (7th ed., p. 1996). Elsevier, with permission.

Show Text Description

Activated charcoal, clay, sand, and gravel

Many household and field filters include granular activated charcoal (GAC), which further treats water by adsorbing organic and inorganic chemicals—including chlorine compounds, iodine compounds, and most heavy metals—thereby improving odor, taste, and safety. GAC filters trap, but do not kill, microorganisms and they are generally not rated for microbe removal. In resource-limited international settings, communities and households might build effective filters made from ceramic clay or simple sand and gravel (i.e., slow sand or biosand).

Chemical disinfection

Halogens

Chlorine compounds and iodine

Chemical disinfectants for drinking water treatment, including chlorine compounds, iodine, and chlorine dioxide, are widely available as commercial products. Sodium hypochlorite, the active ingredient in common household bleach, has been used for over a century and is the primary disinfectant promoted by CDC and the World Health Organization (WHO). Other chlorine-containing compounds, available in granular or tablet formulations (e.g., calcium hypochlorite and sodium dichloroisocyanurate), can also be effective for water disinfection.

An advantage of chemical water disinfection methods is that dosing based on water volume enables their use by individual travelers, small or large groups, or communities. In emergency situations or when other commercial chemical disinfection

water treatment products are not available, household bleach can be used with dosing based on water volume and clarity.

Given adequate concentrations and length of exposure (i.e., contact time), chlorine and iodine have similar activity and are effective against bacteria and viruses. Although *Giardia* cysts are more resistant than bacteria and viruses to chemical disinfection, field-level concentrations of chlorine and iodine are effective against this parasite when longer contact times are used.

Another common protozoan parasite, *Cryptosporidium*, is poorly inactivated by chlorine- or iodine-based disinfection at practical concentrations, even with extended contact times. Chemical disinfection can be supplemented with filtration to remove these oocysts from drinking water.

Cloudy water contains substances that neutralize chemical disinfectants and requires higher concentrations of

or longer contact times with chemical disinfectants. To improve taste as well as effectiveness of disinfection, clarify cloudy water using settling, C-F, or filtration (described above) before adding the disinfectant.

Because iodine has physiologic activity, WHO recommends limiting drinking iodine-disinfected water to a few weeks. People with unstable thyroid disease or known iodine allergy should not use iodine for chemical disinfection. In addition, pregnant women should not use iodine to disinfect water for prolonged periods of time because of potential adverse effects on the fetal thyroid. Advise pregnant travelers to use an alternative method of water disinfection (e.g., heat, chlorination, filtration).

Taste preference for iodine over chlorine is individual; neither is particularly pleasant in doses recommended for field use. The taste of halogen-treated water can be improved by running the

water through a filter containing GAC, by adding a pinch of powdered ascorbic acid (vitamin C), or by adding 5–10 drops of 3% hydrogen peroxide per quart (1 L; 32 oz) of water, then stirring or shaking, which can be repeated until the taste of chlorine or iodine is gone.

Chlorine dioxide

Chlorine dioxide (ClO_2) kills waterborne pathogens, including *Cryptosporidium* oocysts, at practical doses and contact times. Several commercial ClO_2 products are available in liquid or tablet form, but relatively few data are available on the use of these products for different water conditions.

Silver and other products

Silver ion has bactericidal effects in low doses and attractive features including lack of color, taste, and odor. Disadvantages are that silver ion concentration in water can be strongly affected by adsorption onto the surface of the container and that limited testing

on viruses and cysts has been performed. Silver is widely used by European travelers as a drinking water disinfectant, but in the United States, silver is approved only for maintaining microbiologic quality of stored water. Silver is available alone or in an effective combination with chlorine in tablet formulation.

Several other common products, including hydrogen peroxide, citrus juice, and potassium permanganate, have antibacterial effects in water and are marketed in commercial products for travelers. However, none has sufficient data to recommend them for water disinfection at low doses used in the field.

Ultraviolet irradiation

Ultraviolet radiation (UVR) kills bacteria, viruses, and both *Giardia* and *Cryptosporidium* oocysts in water. Efficacy depends on dose and exposure time. In the field, portable battery-operated units capable of delivering a metered, timed dose of UVR are an effective way to disinfect 1–2 L

(or 1–2 qt) of clear water at a time. Because suspended particles can shield microorganisms from UVR, UV irradiation units have limited effectiveness in disinfecting water with high levels of suspended solids and turbidity. Larger units with water flowing through fixed UVR sources are available for use where a power source is available.

Solar irradiation

Using UVR from sunlight to irradiate water (i.e., solar disinfection or SODIS) can improve the microbiologic quality of water and can be used in austere emergency situations. Because natural UVR is also blocked by particles, travelers should clarify highly turbid water first. The optimal procedure is to use transparent bottles (e.g., clear plastic beverage bottles) laid on their side and exposed to sunlight for a minimum of 6 hours with intermittent agitation. Under cloudy weather conditions, water must be placed in the sunlight for

2 consecutive days. The Swiss Federal Institute of Aquatic Sciences and Technology provides more details on SODIS.

Choosing a treatment or disinfection technique

Table 1.8.1 summarizes advantages and disadvantages of field water disinfection techniques, and Table 1.8.3 provides effective techniques based on water source. Travelers can use a UVR-generating device or liquid bleach (1–2 drops per L [qt] of water) to disinfect tap water. Trekkers or campers might prefer to use filters rated to remove viruses or chlorine dioxide. Only reverse osmosis can desalinate water for ocean travel. Travelers should practice using treatment or disinfection methods before leaving for their destination.

While using appropriate water treatment and disinfection methods can help protect travelers from waterborne diseases, travelers should be

cognizant that prevention of diarrheal illness and other infections is promoted by the combination of using safe drinking water, proper hygiene, and adequate sanitation.

Table 1.8.3: Water treatment and disinfection techniques depending on water source

	Primary concern	

	Ciardia, enteribacteria	

	Bacteria, *Giardia*, some viruses	

	Allente	1.

	ricpathogens, including *Cryptosporidium*	2.

	Allente	

	ricpathogensplusposttaste,snell	

	Allnicroorganisnsaswellasallt	

Notes

Abbreviations: CF: Coagulation-flocculation; UVR: Ultraviolet radiation; WASH: Water, sanitation, hygiene

[a]In resource-poor situations, techniques may include solar disinfection SODIS, solar pasteurization with a solar oven, locally available source of chlorine (household bleach), or locally fabricated filters (see references)

[b]Includes heat, UVR, filtration (microfilter), or halogens

[c]Includes agricultural run-off with cattle grazing or sewage treatment effluent from upstream villages or towns